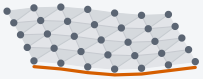
Screened multiview finite-element identification

Geometry and motion → hybrid prior → screened FEM → audited material field

1 Geometry and constraints



surface mesh • tetrahedra
boundary • contacts

$\Pi_v, \mathbf{f}$

project + load

2 Multiview motion

4 loads • 3 views
7-frame tracks + force

$\mathbf{z}_{i\ell vt}$

tokenize

3 Response encoding

visibility mask
temporal response tokens

θ_0

MeshGNN prior

4 Hybrid material initializer

MeshGNN field + calibrated radius

$\theta_0 = (\hat{E}_b, \hat{\rho}_E, \hat{\mathbf{c}}, \hat{R})$

ΔJ

screen

5 Physics decision

seven probes
retain prior or bounded MAP

accept / roll back

audit

6 Audited output

✓

accept or roll back
field • interval • diagnostics

Learned-prior initializer + support-screened MAP FEM + objective-based fallback

Selection protocol

geometry split • train fit • validation selection • locked test